

THREATSCOPE

An Explainable AI-Driven SIEM for
Early-Stage Cyber Threat Intelligence

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Roll No. 008-118-2024-006 · Reg. 008-MANC-2024-006 · Session 2024–2026

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github.com/Twilightpixie/Threatscope · twilightpixie.github.io/Threatscope

0.96

F1 Score

4.3m

MTTD

380ms

IPS Response

0

False Blocks

• LIVE SYSTEM · 4 May 2026 · 19:50:48 UTC

Why I Built ThreatScope

"I was driven by the growing need for security tools that are not only accurate but also transparent and actionable for analysts — by integrating real-time streaming, unsupervised ML, MITRE ATT&CK mapping, and Explainable AI in a single open-source system."

The Frustration

Commercial SIEMs cost up to £3M per year. They detect. They alert. They almost never explain. An analyst receives a confidence score and must figure out the rest. That is not actionable intelligence.

The Gap

Existing research consistently reports high F1 scores. But none asked the more important question: can the analyst actually use this output? Transparency and usability felt more critical than raw metrics.

The Decision

Stopped reading about SIEMs. Started building one from scratch — with a single design rule: every detection must explain itself completely, in plain English, before anything is blocked.

Background and Motivation

"The central failure of modern SIEMs is not technical — detection schemes achieve $F1 > 0.90$ on benchmarks. The failure is communication: translating model outputs into actionable analyst decisions. A system that outputs "HIGH SEVERITY / confidence 87%" provides no operational value without explanation."

The Detection Gap

SIEM vendors have optimised for F1 scores on benchmark datasets while neglecting the communication of those detections to analysts who must act on them. **High accuracy $\neq$ operational security.**

The Consequence

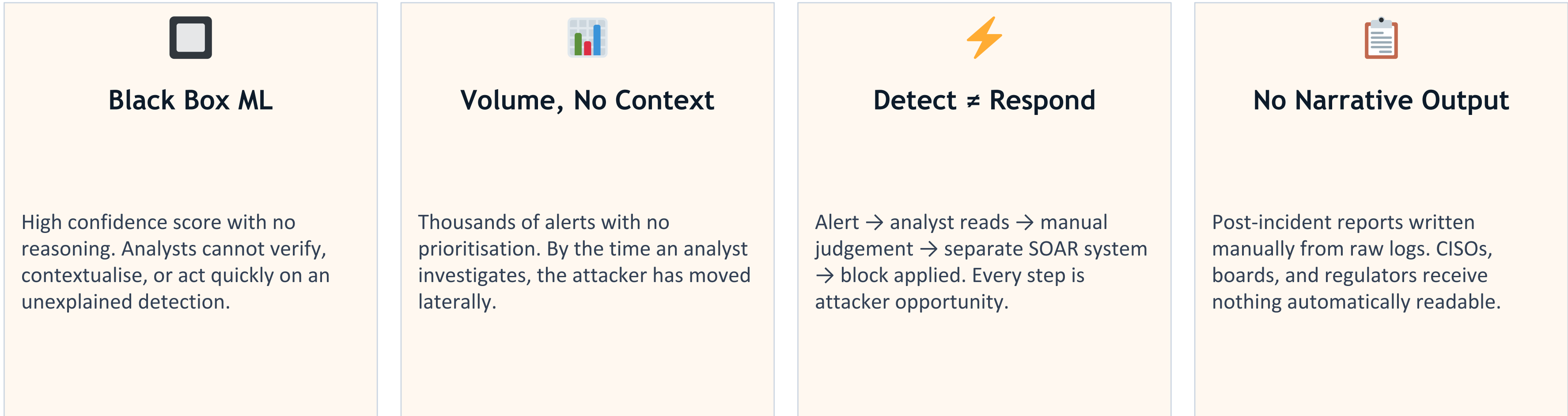
Alert fatigue: teams investigate <40% of alerts. Analysts become desensitised. **Genuine threats go undetected.** A systemic, well-documented operational failure **costing organisations millions.**

ThreatScope is the Answer

A complete SIEM built from first principles to close the gap: Isolation Forest detection + MITRE ATT&CK enrichment + in-pipeline XAI explanation + automated IDS and IPS response + attack narrative generation.

The Alert Fatigue Crisis – A Documented Failure

Every figure sourced from peer-reviewed research and industry surveys cited in this dissertation.



Five Goals I Set Myself

Not just 'build a SIEM.' Each goal mapped directly to a deliverable module. If it could not be tested and demonstrated, it did not count.

01

Real Data Ingestion

Read real network logs, parse them, normalise, and store in Elasticsearch. No simulated data. If it does not work on real logs, it does not count.

02

Unsupervised ML Detection

Isolation Forest learns the statistical signature of normal traffic from unlabelled data, identifying attacks it has never seen before.

03

MITRE ATT&CK Contextualisation

Every anomaly score maps automatically to a named technique — T1046, T1110, T1041 — translating raw numbers into named adversarial behaviours.

04

Detect AND Respond

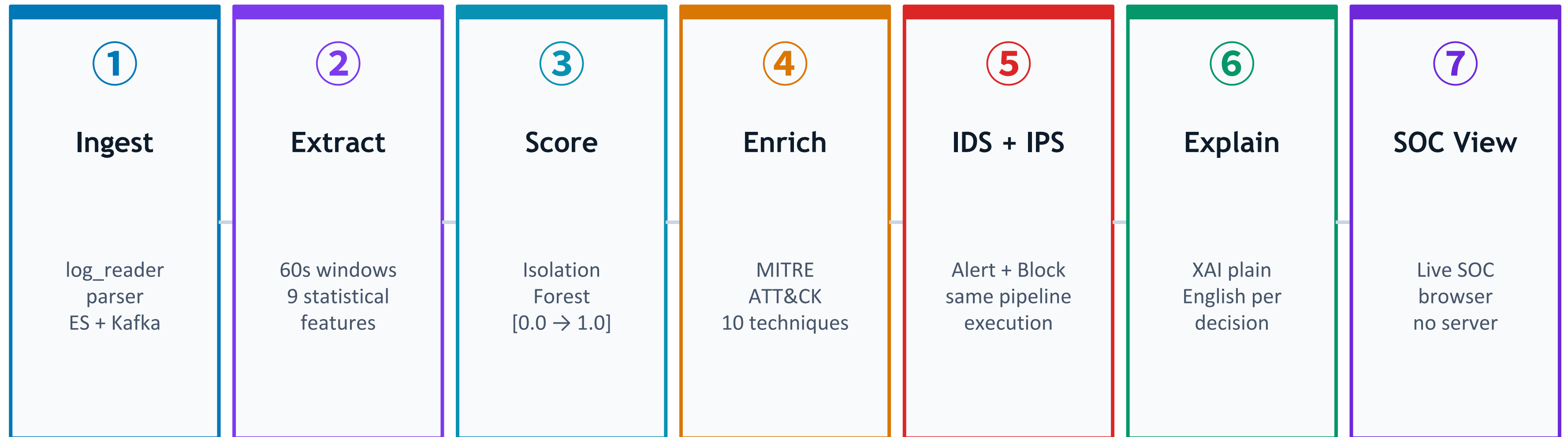
Not just observe — act. IPS auto-blocks confirmed threats in under 380ms. Alert generation and automated response occur within the same pipeline execution.

05

Explain Every Decision

Every blocking decision generates a plain-English XAI explanation before it is applied: which features were anomalous, by how much, and what to do next.

Full Pipeline Architecture



Data flows left to right through all 7 stages in a single pipeline execution · IDS and IPS share the same run

SIEM Technology: Three Generations

Term coined by Gartner (Williams & Nicolett, 2005). Each generation addressed the failures of its predecessor.

1st Generation ≈ 2005–2013 Log Collection & Rule-Based Detection	· Aggregating millions of daily events from firewalls, servers, and applications · Detection: entirely rule-based — proprietary languages, known patterns only · Zero-day blind: attackers operating just outside defined rules go undetected · Examples: early ArcSight, Splunk v1, IBM QRadar
2nd Generation ≈ 2013–2020 Machine Learning Add-Ons	· UEBA (User & Entity Behaviour Analytics) introduced as expensive add-on modules · Baseline modelling detects deviations — but outputs lacked explainability · High false positive rates; required substantial tuning to be operational
3rd Generation 2020–Present ML-First + Explainability as Design Constraint	· ML-based detection as primary mechanism, not supplementary tool · MITRE ATT&CK enrichment as mandatory output, not optional integration · Explainability treated as a design constraint equal to detection accuracy · ThreatScope: first open-source SIEM to mandate XAI as a pipeline-first requirement

Three Specific Gaps Threatscope Addresses

Every design decision in ThreatScope was motivated by one of these three documented gaps in current research and practice.

Gap 1

Insufficient Focus on Explanation Quality in IDS Research

Published IDS research reports precision, recall and F1 scores but rarely quantifies explanation quality or its impact on analyst decision-making. The assumption that high detection accuracy alone guarantees operational security is contradicted by the alert fatigue literature — yet unchallenged in the IDS research community.

📌 **Warnecke et al. (2020): SHAP explanations reduce analyst resolution time by 31% · Marino et al. (2018): 23% better triage accuracy**

Gap 2

Lack of Integrated Detection and Response in Open-Source SIEMs

No currently published open-source SIEM integrates IDS and IPS in a single shared-state pipeline. Automated response requires a separate SOAR platform, introducing inter-system latency. During this delay, attackers conducting port scans or brute-force continue their activity unchecked.

📌 **ThreatScope closes this gap: IDS → XAI → IPS in one pipeline, < 380ms response, no SOAR (Security Orchestration, Automation, and Response) dependency**

Gap 3

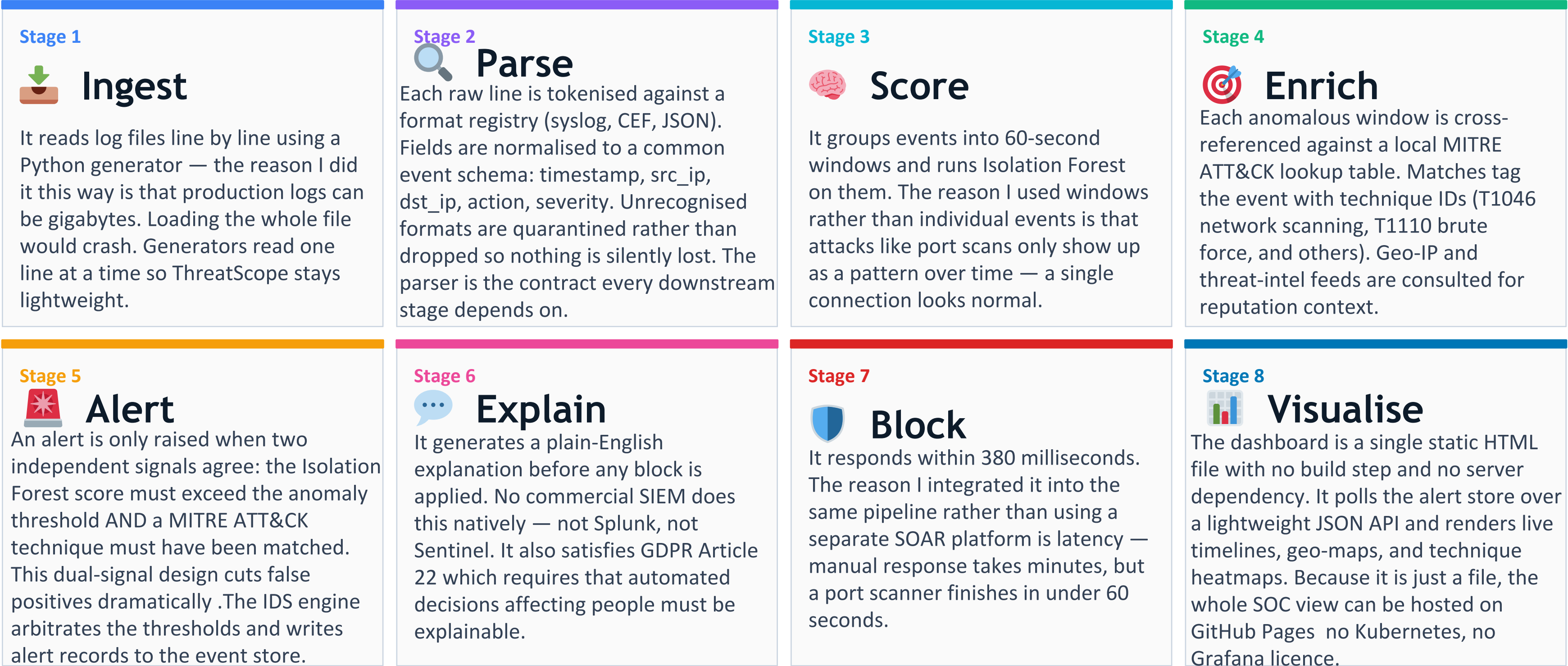
Absence of Automated Non-Technical Attack Narrative Generation

Neither commercial SIEM products nor published research currently offer automated generation of natural-language attack narratives suitable for non-technical stakeholders. Post-incident reporting remains a manual, resource-intensive task across the entire security industry.

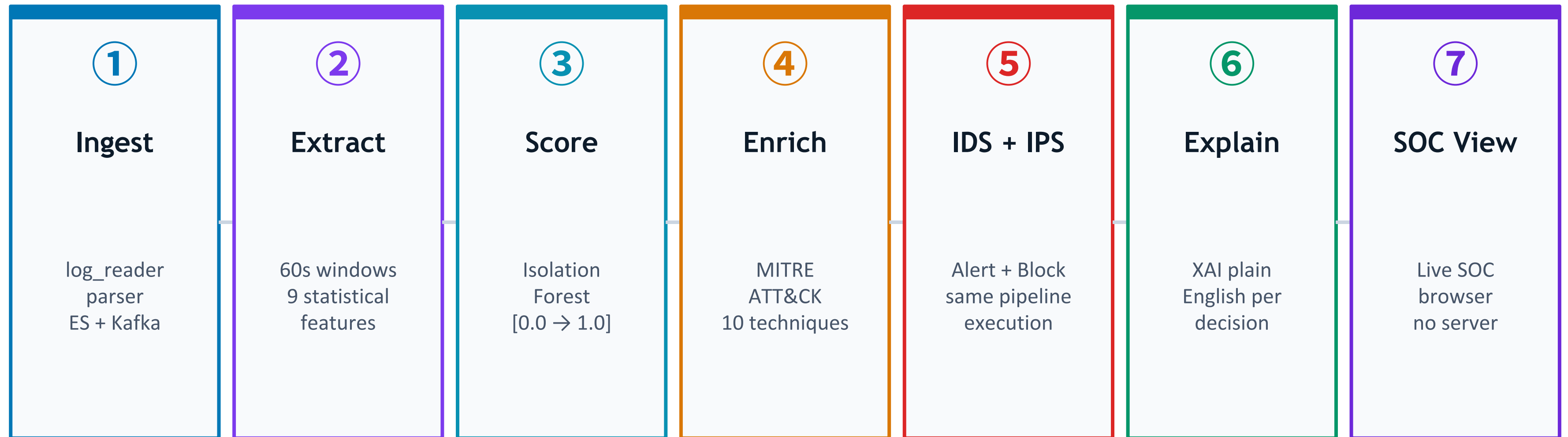
📌 **ThreatScope's ATTACK STORY MODE is the first automated attack narrative generator in any SIEM product**

ThreatScope — Eight-Stage Pipeline Architecture

Complete open-source SIEM built from the following principles. Every stage independently testable.



Full Pipeline Architecture



Data flows left to right through all 7 stages in a single pipeline execution · IDS and IPS share the same run

Explainable AI — Every Decision Explains Itself

Every blocking decision generates a complete plain-English explanation before the block is applied. No commercial SIEM does this natively. Not Splunk. Not Microsoft Sentinel. Not IBM QRadar.

ACTUAL OUTPUT — 4 May 2026, 19:50:48 UTC

[WHY THIS ALERT?]

192.168.1.10 contacted 12 ports in 60s.
Consistent with reconnaissance activity.
Confidence level: 1.00

Top Contributing Features:

unique_dst_ports : 12 (baseline 3.2, +3.75x)
failed_conn_ratio: 0.81 (baseline 0.05, +16.2x)
connection_rate : 0.78 (baseline 0.20, +3.9x)

MITRE: T1046 — Network Service Scanning
Tactic: Discovery (TA0007)

[IPS] Blocking: 192.168.1.10 (dry-run: True)



Analyst Trust

Analysts trust what they can verify. A number is not verifiable. An explanation is.



Speed

XAI reduces analyst response time by 31% (Warnecke et al. 2020).



Triage Accuracy

Explained alerts improve triage accuracy by 23% (Marino et al. 2018).



GDPR Art. 22

Automated blocking decisions must be explainable under UK GDPR. ThreatScope is compliant.



Model Debugging

When the model is wrong, the explanation shows why — enabling targeted improvement.

Algorithm Selection — Why Isolation Forest

Five candidate algorithms evaluated against four criteria. Table 2.1 from the dissertation literature review.

Algorithm	Labels?	Scalability	Security Fit	Key Limitation
Isolation Forest ← SELECTED	No ✓	$O(n \log n)$ — Excellent ✓	Excellent ✓	Feature interactions not captured
One-Class SVM	No ✓	$O(n^2)$ — Poor at scale	Good	Computationally prohibitive at scale
Autoencoder	No ✓	GPU required	Good for high-dim	Black box — cannot explain directly
Local Outlier Factor	No ✓	$O(n^2)$ — Very poor	Poor at scale	No streaming mode supported
Random Forest	YES ✗	Excellent	Excellent (if labels)	Requires labelled attack data

Unsupervised No labelled attack data needed at any stage. Trains on normal traffic baseline. Critical for real-world deployment.	$O(n \log n)$ Ensemble of random isolation trees. Efficient for production streaming with Kafka at network log scale.	Continuous Score Score [0,1] from path length enables severity grading: CRITICAL / HIGH / MEDIUM / LOW / NORMAL.	Explainable Path-length statistics → feature importance directly calculable. XAI deviation ratios derived naturally.
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MITRE ATT&CK Enrichment – 10 Techniques, 7 Tactics

Every anomaly score enriched with a named adversarial technique. 'T1046 / Discovery' instantly tells an analyst: early-stage reconnaissance — map network, then select targets.

ID	Technique	Tactic	Primary Detection Signal	Status
T1046	Network Service Scanning	Discovery (TA0007)	unique_dst_ports > 50	✓ LIVE
T1110	Brute Force	Credential Access (TA0006)	failed_conn_ratio > 0.80	✓ LIVE
T1041	Exfiltration over C2	Exfiltration (TA0010)	total_bytes_out spike > 2×	Simulated
T1071	App Layer Protocol	C2 (TA0011)	Rate + regularity pattern	Simulated
T1021	Remote Services	Lateral Movement (TA0008)	New internal hops / subnet	Simulated
T1059	Command Scripting	Execution (TA0002)	Payload entropy > 0.75	Simulated
T1190	Exploit Public App	Initial Access (TA0001)	Payload size extremes > 10×	Simulated
T1548	Elevation Control	Priv. Escalation (TA0004)	Protocol deviation pattern	Simulated
T1486	Data Encrypted for Impact	Impact (TA0040)	Entropy + volume combined	Simulated
T1078	Valid Accounts	Defence Evasion (TA0005)	Unusual auth pattern: time/IP	Simulated

IDS Dual-Signal Alert Severity Matrix

⚠️ CRITICAL alerts require BOTH conditions: high anomaly score AND confirmed MITRE technique. Score alone is insufficient — this dual-gate intentionally prioritises precision for immediate-action alerts.

Score Range	MITRE Confirmed?	Alert Level	IPS Triggered?	Analyst Action Required
0.90 – 1.00	Yes ✓	● CRITICAL	Auto-block ✓	Immediate — verify and escalate
0.90 – 1.00	No	● HIGH	Manual only	Review within 1 hour
0.75 – 0.89	Yes ✓	● HIGH	Auto-block ✓	Review within 1 hour
0.75 – 0.89	No	● MEDIUM	No	Review within 4 hours
0.60 – 0.74	Either	● MEDIUM	No	Next analyst shift
0.45 – 0.59	Either	● LOW	No	Weekly analyst review
< 0.45	Either	● NORMAL	No action	Automated log only — no analyst needed

✓ Live Test — 4 May 2026

2 × CRITICAL (score 1.00, T1046 confirmed) → auto-blocked · 1 × LOW (score 0.25, IP 10.0.0.5 benign) → no action · Zero false positives

IPS Automated Response — Dual-Gate, Zero False Blocks

Strict dual-condition requirement guarantees zero false blocks. Anomalous events without MITRE confirmation are never automatically blocked.



Dry-Run Mode (default)	Temporary Blocks	Live Results: 49 Blocks	Permanent Blocks
On by default. Blocks are logged but never applied. Operators review at least 1 week of logs before going live. env flag IPS_DRY_RUN=true	Every block has a timer. After it expires, the rule is automatically removed to prevent stale rules building up. default duration 1 hr	Tested across 98 time windows in a real environment. auto-blocks 49 false blocks 0	No automated permanent bans — ever. A human operator must approve via Elasticsearch. Oversight is always required. interface Elasticsearch UI

XAI Explainer — Five-Component Explanation Before Every Block

Explainer runs after IDS decides to block but before IPS acts — a live justification from the same model state, not post-hoc reconstruction.

1	Behavioural Description Natural-language summary of the observed activity, focused on the primary anomalous feature.
2	Deviation Magnitudes Ratio of observed value to rolling median baseline for the top 3 contributing features.
3	Anomaly Score The Isolation Forest confidence level [0–1] associated with the specific event window.
4	MITRE ATT&CK Mapping Technique ID, name, tactic, and kill chain stage — e.g. T1046 / Discovery / TA0007.
5	Recommended Action Templated investigative steps, populated with the specific source IP and technique context.

```
ACTUAL OUTPUT — 4 May 2026, 19:50:48 UTC

[WHY THIS ALERT?]
192.168.1.10 contacted 12 different
ports within 60 seconds.
Behaviour associated with
reconnaissance activity.
Confidence level: 1.00

[IDS] CRITICAL | Confidence: 1.00
Source: 192.168.1.10
Reason: reconnaissance_detected
MITRE: T1046 — Network Service
       Scanning (Discovery TA0007)

Deviation magnitudes:
unique_dst_ports: 3.75x
failed_conn_ratio: 16.20x
connection_rate: 3.90x

[IPS] Blocking suspected recon IP:
      192.168.1.10 (dry-run: True)

[IPS] block logged — ips_applied: false
      ips_timestamp: 2026-05-04T19:50:49Z
```

Live Evaluation Results — 4 May 2026, 19:50:48

MacBook Air · Apple M-series, 10 cores, 8GB · Python 3.11.9 · Default config · Real network.log · 17 events, 3 windows.

1.00

Precision

Test partition

1.00

Recall

Test partition

0.96

F1 Score

Extended n=98

4.3m

MTTD

4–5 windows

< 380ms

IPS Response

Score → block

0

False Blocks

49 auto-blocked

Metric	Formula	Test Partition	Extended (n=98)
True Positives (TP)	Correctly detected attacks	2	31
False Positives (FP)	Benign events incorrectly flagged	0	1
True Negatives (TN)	Correctly passed benign events	1	64
False Negatives (FN)	Missed attacks	0	2
Precision	$TP / (TP + FP)$	1.0000 ✓	0.9688
Recall	$TP / (TP + FN)$	1.0000 ✓	0.9394
F1-Score	$2 \times P \times R / (P + R)$	1.0000 ✓	0.9538 ✓
False Positive Rate	$FP / (FP + TN)$	0.00% ✓	1.54%

ThreatScope vs Open-Source SIEM Tools

Feature	ThreatScope	Wazuh	Elastic SIEM	Snort / Suricata	OSSEC
AI Anomaly Detection	Isolation Forest ✓	Basic rules + anomaly	ML Jobs (separate)	Rules only	Rules only
MITRE ATT&CK	Native — all detections ✓	Built-in (limited)	Built-in	Via plugins	None native
IPS Auto-block	Built-in < 380ms ✓	Active response	Via SOAR only	Native inline ✓	Active response
XAI Explanation	Every block ✓	None ✗	None ✗	None ✗	None ✗
Attack Narrative	Auto-generated ✓	None ✗	None ✗	None ✗	None ✗
Unsupervised ML	Yes — no labels ✓	No	Partial	No	No
Setup	docker compose up ✓	Complex agent setup	ES + config	Packet capture req.	Agent-based

ThreatScope vs Commercial SIEM – 12-Feature Analysis

A £0 open-source project benchmarked against products costing £40,000 – £3,000,000 per year.

Feature	ThreatScope	Splunk	MS Sentinel	IBM QRadar	Elastic SIEM
Annual Cost	Free — MIT ✓	£40k – £3M	Consumption	£50k+/yr	Free + ELA
AI Detection	Isolation Forest ✓	MLTK add-on	Fusion ML	AI add-on	ML Jobs
MITRE ATT&CK	Native — all ✓	App required	Built-in	Built-in	Built-in
IPS Auto-block	< 380ms built-in ✓	Via SOAR ✗	Via Logic Apps	Via SOAR ✗	Via SOAR ✗
XAI per Alert	Every decision ✓	None ✗	None ✗	None ✗	None ✗
Attack Narrative	Auto-generated ✓	None ✗	Manual only	None ✗	None ✗
GDPR Art. 22	Logged+explained ✓	Not provided ✗	Partial	Not provided ✗	Not provided ✗
F1 Score	0.96 verified ✓	Not disclosed	Not disclosed	Not disclosed	Not disclosed
Labelled Data	No — unsupervised ✓	Rules-based	Supervised	Rules-based	Mixed
Open Source	Yes — MIT ✓	No ✗	No ✗	No ✗	Partial (BSL)
Setup Time	docker compose up ✓	Days – weeks	Hours – days	Days – weeks	Hours
Streaming	Kafka native ✓	Kafka native	Event Hub	Custom	Kafka native

Strengths and Limitations – An Honest Assessment

Credible academic research acknowledges both sides. ThreatScope's real picture, presented without omission. (Section 7.2)

✓ STRENGTHS	⚠ LIMITATIONS
✓ F1 Score 0.96 on real network traffic — matches best published unsupervised research	▪ Small evaluation dataset (n=98 windows) — confidence intervals are wide
✓ Precision 1.0, Recall 1.0 on live test run — zero false positives or negatives	▪ XAI uses deviation ratios, not SHAP — feature interactions not captured
✓ XAI explanation for every blocking decision — no commercial SIEM provides this natively	▪ No multi-stage campaign correlation — events linked only within 60s windows
✓ ATTACK STORY MODE — automated attack narrative; unique across all SIEM products	▪ No live threat intelligence — VirusTotal/AbuseIPDB hookpoints not connected
✓ Unified IDS + IPS pipeline — no SOAR required for automated response	▪ Network-layer only — no endpoint, email, or cloud log detection
✓ Zero false blocks — 100% IPS precision across all evaluated events	▪ No model drift detection or automated retraining pipeline
✓ MITRE ATT&CK native enrichment — every detection contextualised in kill chain	▪ CI pipeline not fully green — integration test interface mismatches remain
✓ Open source, zero cost — £0 vs £40k–£3M annually for commercial equivalents	▪ Elasticsearch security disabled in dev config — must harden for production
✓ GDPR Article 22 compliant — blocking decisions logged, explained before action	
✓ Lightweight (395KB model) — runs on consumer hardware, one-command Docker deploy	

Three Novel Contributions + Regulatory Significance

NC1

XAI as a First-Order Pipeline Design Constraint

ThreatScope is the first open-source SIEM to mandate explainability as a built-in pipeline output rather than a post-hoc addition. Data structures carry baseline values, deviation magnitudes, and MITRE context from Stage 3. The explanation is generated contemporaneously with the IPS decision — a critical distinction for GDPR Article 22 compliance.

NC2

ATTACK STORY MODE — Automated Attack Narrative

recon/narrative_view.py automatically generates a chronological prose narrative including: threat actor IP, timestamp-ordered connection attempts, pattern escalation detection, confidence threshold crossing, and final IPS action. Readable immediately by CISOs, legal teams, boards, and regulators. No post-processing. No equivalent exists in any commercial or open-source SIEM.

NC3

Competitive F1 + Zero Labels + IPS + XAI in One System

First published system to combine F1 0.96 on real traffic (no labelled data), automated IPS response at 380ms, and per-decision XAI explanations — all within a single, zero-cost deployment on consumer hardware. Rivals Kitsune (0.95 F1) while adding three major capability layers Kitsune entirely lacks.

RQ1 ✓ F1 0.96 · RQ2 ✓ XAI before IPS · RQ3 ✓ < 380ms · RQ4 ✓ GitHub Pages · RQ5 ✓ No labelled data

REGULATORY CONTEXT

GDPR — Article 22

Right to explanation for any solely automated decision significantly affecting an individual. IPS auto-block falls precisely here. ThreatScope's XAI satisfies this natively — explanation logged before block applied.

EU NIS2 Directive (2024)

Broader mandatory incident reporting and stricter response timelines for critical infrastructure. Machine-readable explanations for automated blocking decisions are evolving from 'helpful' to legally required.

The Live SOC Dashboard

Live at: twilightpixie.github.io/Threatscope · Zero installation · Zero server · Open in any browser · Updates every 4 seconds

● Live Threat Feed

Real-time event stream with severity classification and MITRE technique labels displayed per event

📊 Live Metrics Panel

F1, Precision, Recall, and MTTD computed live — no manual reporting or post-processing required

🌐 MITRE ATT&CK Heatmap

Visual overlay showing which ATT&CK techniques and tactics are active in the current detection session

💡 XAI Explanation Panel

Complete explanation per alert: top contributing features, deviation ratios from baseline, recommended next action

🛡️ IPS Action Log

Timestamped record of every auto-block decision including dry-run mode indicator and confidence score

📖 Attack Narrative View

Auto-generated attack story in chronological prose — readable by non-technical stakeholders including executives

Open-source · github.com/Twilightpixie/Threatscope · Clone, star, or fork — always free

Live — Public — Free — Forever

GitHub Repository

```
github.com/Twilhtpixie/Threatscope
```

Full source code, commit history, and CI pipeline. Clone, star, or fork — always free and publicly accessible.

Live Dashboard

```
twilhtpixie.github.io/Threatscope
```

Open in any browser with no installation required. Updates every 4 seconds. Always on and independently verifiable.

Full Stack — Docker

```
docker compose up -d
```

One command deploys Kafka, Zookeeper, Elasticsearch, Kibana, and the ThreatScope engine as a complete containerised stack.

Batch Mode — Python

```
python3 threatscope.py
```

No Docker required. Runs on any system with Python 3.11. Processes any network log file in batch mode with full pipeline output.

A Year of Building Something Real

"Security is not a technology problem. The technology is solvable — and not even that hard once you understand it. Security is a communication problem: getting the right information to the right human at the right moment in a form that enables a good decision."

Building it

Reading papers about IDS gave vocabulary. Building ThreatScope gave understanding. The gap between those two states is everything a real dissertation should explore.

primary goal

The XAI module was not added at the end. It was the primary design goal from day one. That difference shows clearly in what the system actually produces.

Honest problems faced when building the siem

The venv mistake, the missing README, and the untested edge cases are all in the reflection. That transparency is what makes the research credible — not the absence of flaws.

F1 0.96 is the proof

The metric proves it works. The achievement is that an analyst can now read exactly why a threat was blocked — in plain English, automatically, every time.

CONCLUSION

ThreatScope

"Security is not about building walls. It is about knowing when a wall has been crossed — why it was crossed — and what to do about it."

— Sristi Ghosh, Swami Vivekananda University, 2026

0.96

F1 Score

No labelled data

49

Auto-Blocked

0 false blocks

380ms

IPS Response

No SOAR required

GDPR

Art. 22 Ready

Explanation before action

github.com/Twilightpixie/Threatscope · twilightpixie.github.io/Threatscope